



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2023-24/EMD/LAB /540318

Date:

05/10/2023(10:18:31)

Subject:REVISION OF LMI/OGN/OPS/ELEC/020 CLEAN CONDITIONS
DURING MAJOR OVERHAULING OF GENERATORS.

**Subject: REVISION OF LMI/OGN/OPS/ELEC/020 CLEAN CONDITIONS DURING MAJOR
OVERHAULING OF GENERATORS.**

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

No changes envisaged by COS. Only periodic reviewing done after two years. No changes in the document.

May please approve.

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Note No:2

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Note No:4

Updated

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Note No:6

CHECK LIST ATTACHED.

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LMI reviewed and found in Order.

May be approved

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Note No:11

Approval my please be accorded for the new LMI.

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Note No:13

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On:19/10/2023(11:25:11) **Ref:**SMTPP/ELECT MAINT/2023-24/EMD/LAB /540318

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION
LMI/OGN/OPS/ELEC/020
ISSUE NO.: 1/ REVISION; 2, DATE: Oct 2023

Clean conditions during major overhauling of Generators

APPROVED FOR

IMPLEMENTATION By _ -----
GROUP GENERAL MANAGER (SMTTP)

DATE -----

NTPC LIMITED

SIMHADRI SUPER THERMAL POWER STATION

OPERATION AND MAINTENANCE DEPARTMENT

LMI(Name) :	Clean conditions during major overhauling of Generators		
LMI No.	LMI/OGN/OPS/ELEC/020		
ISSUE NO.	1	Revision : 2	Date: Oct 2023

	DESIGNATION	SIGNATURE & NAME
Prepared by	Sr. Manager (EM)	
Checked by	DGM (EM)	
Checked by	AGM (EM)	
Reviewed by	AGM (EM) I/C	
Reviewed by	GM (Maint)	
Reviewed by	GM(O&M)	

Clean conditions during major overhauling of Generators

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1.0 OBJECTIVE:

There are instances of serious damages caused to generators as a result of foreign material entering or being left in generators during overhauling. This LMI Provides guidelines for maintenance of clean conditions during working of Generators. This LMI also provides guidelines or measures to be taken to prevent ingress of foreign materials into Generators and their associated systems during dismantling, working and Re-assembly.

2.0 SCOPE:

This is applicable to Unit # 1, 2, 3 & 4, Generator 500MW of Simhadri.

3.0 REFERENCE DOCUMENTS:

OGN/OPS/ELEC/020.

4.0 SUPERSEDED DOCUMENTS:

LMI/OGN/OPS/ELEC/020, ISSUE NO.: 1/ REVISION; 1, DATE: Jan 2022

5.0 SPECIFIC INSTRUCTIONS TO BE FOLLOWED:

The clean conditions during work of Generators shall be carried out in all the units of Simhadri as per 6.0 of this LMI.

6.0 METHODOLOGY OF IMPLEMENTATION:

Whenever generator end covers are opened and any work is to be carried out inside, the following precautions will be taken:

6.1 Preparation:

- a) **Barriers:** A secure personnel barrier shall be erected around the Generator at the operating Floor level.
- b) **Working Area:** The area within the barrier shall be thoroughly cleaned before any part of the Generator is dismantled. The clean conditions shall be maintained throughout the working period. Dismantled PMG stator, main exciter unit and threaded out Generator rotor would be wrapped with

polythene covers to avoid ingress of dust and moisture into them. Work area would be marked for storage of major dismantled parts of Generator.

C) Protection from the Top: A canopy/cover shall be provided to protect the generator due to falling of objects and any work over the Turbo Generator would be avoided. The EOT crane shall be inspected for loose debris and cleaned if necessary.

d) Access to Clean Condition Area:

- There shall be only one access point into the Generator clean conditions area, preferably at the exciter end. This point shall be manned continuously or locked shut when no work is in progress.
- A small floor mat at the entrance shall be kept to collect small debris. These mats shall be replaced with cleaned ones daily.
- A bin will be maintained outside clean area to dump the waste material.
- The Following Precautions shall be followed by the Person entering the Generator:
 - ❖ Maintain a check list of all materials taken into and out of the area.
 - ❖ Maintain a check list of all tools and special equipment taken into and out of the area.
 - ❖ Refuse entry to all personnel except those holding a pass issued by EIC.
 - ❖ Ensure all persons entering have clean condition overalls and overshoes and they carry on personal effects likely to drop out of pockets etc. Loose items such as coins, keys, pen, pencils etc. should be emptied into a small bin before entering the clean area.
 - ❖ Ensure overshoes are worn by personnel working inside the generator frame.
- Contractor's Responsibilities: It must be made clear to contractors when a contract is placed that they will be expected to provide clean conditions overalls, shoes, treadmats, bins etc. suitably marked.

e) Foreign Bodies:

- To minimize the risk of a foreign body entering any system, the following should be ensured:
- The station staff shall attempt to insist to all men of supervisory status and workmen, the importance of keeping systems internally clean. The methods used should include lectures, written information and personal discussion.
- Station staff and contractor's employees should be encouraged to inform their superiors of any tool or item that may have been dropped into the generator, mislaid or not accounted for in the clean conditions area. It is better to identify, locate and remove the item even at the cost of extended outage.
- Special attention should be given to any magnetic material / powder taken inside the clean condition area to avoid its ingress into the generator.

f) Use of covers and Blanks:

- Temporary end covers should be fitted to the generator and only removed whilst work is being carried out. Windings shall be kept warm and dry.
- All disconnected pipe work should be fitted with purpose-made blanks or plugs which do not allow re-assembly whilst in position.
- All temporary covers, blanks and shrouds should be clearly labelled or distinctively identified, e.g., these items may be brightly painted so that they cannot be easily left in position during re-assembly.

6.2 Precautions to be taken during work in the Clean Conditions Area

No mechanical jobs like cutting, welding, chipping and filing etc. should be carried out in the clean conditions area unless most unavoidable for any major modification's jobs under close supervision of senior staff. In that case the following precautions must be undertaken: -

Cutting:

When a pipe is cut care should be exercised to prevent swarf etc. entering the pipe. No pipe should be left open ended, all loose material should be removed and the end inspected before it is covered with the appropriate cover or blanking plug.

Welding

The number of welding rods allowed into the clean conditions area should be counted, and the rod stubs retained to check they are all accounted for.

Work Supervision

All welding, burning and machining operations in the clean conditions area should be continuously supervised by NTPC staff. Availability of firefighting equipment's near the place of work must be ensured before any welding or burning work is under taken.

6.3 Do's and Don'ts

a) Do's

- wear an overall while on job. Use proper gloves during work.
- wear canvas shoes during winding assembly.
- ensure that tools & instruments used are in good working condition & properly calibrated.
- ensure that all tools used are back in tools box and no tools is forgotten inside the stator.
- use protective creams before working with putties and wash hands with soap.
- use vacuum cleaner for cleaning stator.
- ensure covering of openings in stator body when not working to avoid accidental falling of foreign objects into the stator.

b) Don'ts

- Don'ts use any knives without handles for cutting braiding.
- Don't enter the stator with loose objects like, coins, key bunch, metallic ball pen, pencils etc. in pockets.
- Don't use items, the shelf life of which has already expired.

6.4 Special Warning

It has been seen in the past that workers/technicians/ engineers entering inside the generator for inspection/repair/rehabilitation work use shoes with rubber soles as recommended. The synthetic rubber sole shoes should be worn only just before entering the generator. In case these shoes are worn during their walking outside the generator clean area, these are likely

to pick-up metallic chips or other small nails which will get embedded and unless discovered before entry can later get dislodged during walking inside the generator core. Hence, the shoes should be changed and fully inspected before entering the generator and should not be worn other than for entry into the generator.

6.4 Storage Requirements

6.4.1 Components

Components removed from the turbo-generator should be stored in bins within the clean conditions area. All bins should be labelled stating their contents and purpose. Where larger components cannot be put in bins they should be held in a temporary secure storage area.

Components like rotors which need protection against entry of dust, moisture, debris etc. in the ventilation ducts / clearances shall be properly wrapped with waterproof covers.

6.4.2 Waste Material

All waste material and refuse should be put in bins with correctly fitting lids or covers and labelled clearly 'WASTE'. They should not be in the vicinity of the components.

6.4.3 New Components

All new replacement components should be cleaned before they enter the clean conditions area and shall be inspected before fitting.

7.0 FREQUENCY OF CHECKING AND MEASUREMENTS:

During major overhauling.

8.0 RESPONSIBILITY CENTRES FOR IMPLEMENTATION:

Basic responsibility lies with respective Area in-charge. Overall responsibility lies with section head.

9.0 THE EXCEPTION REPORTING & DISPENSATION OF DEVIATIONS:

Any deviation shall be reported to HOD (EM).

10.0 RECORD KEEPING:

Record of man/material going into Generator will be kept in register LMI/E-020 and Record of work will be maintained in Generator Major Overhaul Log book. Above documents will be available with concerned group of Electrical Maintenance.

11.0 REVIEW:

Documents will be reviewed once in two years by HOP.

CHECKLIST DURING OVERHAUL

UNIT:

SL NO	Person/Material entering generator area	Date	In Time	Out Time	Remarks
1					
2					
3					
4					
5					